

Prince Gédéon GUEDJE

AI & Robotics Researcher

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Research Interests

- Multi-Agent Reinforcement Learning · Autonomous Robotics · Cooperative AI · Computer Vision · Safe Reinforcement Learning · Human-Aware Navigation

EDUCATION

- Master 2 in International Computer Science** Sept 2025 - Aug 2026
University of Clermont Auvergne
 - **Thesis** : "Multi-Agent Reinforcement Learning for 3D Urban Environment Mapping under Energy and Communication Constraints"
 - **Scholarship**: Recipient of the **MIAI Excellence Scholarship**
 - **Core Courses**: Reinforcement Learning, Robotics Perception, Multi-Agent Systems, Stochastic Optimization, Autonomous Systems, Big Data, Complexity.
- Master 1 in Computer Science (Artificial Intelligence)** Sept 2024 - Aug 2025
University of Clermont Auvergne
 - **Academic Honors**: Ranked 1st in the class
 - **Core Courses**: Deep Learning, Database foundations, Advanced Data Analysis, Algorithm , Software Engineering
- Bachelor in Computer Engineering** Licence
Institute of Training and Research in Computer Science (IFRI-UAC), Benin
Oct 2020-Dec 2023
 - **Academic Honors**: Ranked 1st in the class
 - **Thesis**: "NextVision: An Intelligent Video Surveillance System Based on Computer Vision and Natural Language Processing"
 - **Leadership**: Head of the University AI-IOT Club

ACADEMIC PUBLICATION

NextVision , An Intelligent Video Surveillance System Based on Computer Vision and Natural Language Processing

Décembre 2024

Springer (ICICT 2024, Royaume-Uni)

Developed a multimodal architecture using **fine-tuned YOLO** for object detection and **ByteTrack** for robust multi-object tracking integrated with NLP for semantic video analysis .

EXPERIENCES

ML R&D Engineer Intern

Marcy l'Etoile

bioMérieux

Mar 2026-Aug 2026

Driving R&D through the integration of SOTA Foundation Models to develop and deploy Generative AI applications for automated data extraction.

AI Developper | Internship

Croix, Lille

L'AGENTX

May 2025-Aug 2025

Designed and deployed multi-agent systems using **CrewAI** and **LangChain**, leveraging LLMs, embeddings, and vector databases for automated **technical documentation** and complex document .

Visiting Instructor

Clermont-Ferrand

HEXAGONE SCHOOL

Feb 2025 (28h)

Delivered a 28-hour specialized course on **Deep Learning** applied to Facial Emotion Recognition for graduate students, covering **Computer Vision- , Transfer Learning**, and **Vision Transformers** .

Junior Data Scientist

Montpellier, France, Remote

WIM - Workinmusic

Novembre 2023 - Aout 2024

- Research of generative audio architectures including **Tacotron**, **VITS**, and **HiFi-GAN** for speech synthesis.
- Developed automated data acquisition pipelines and generative models like MISTRAL for text-to-lyrics synthesis utilizing **LoRA fine-tuning**.

Volunteering

Avenir Vert Solidaire

Implementing **agroecological** aquaponics systems in orphanages to achieve food autonomy

SKILLS

Artificial Intelligence

Pytorch, OpenCV, Transformers,
Stable Baselines3, Gym, LangChain,
Vision Transformers

Robotics

Ros2 Jazzy, Gazebo, Onshape

Languages de programmation

Python, C/C++, Go

Devops

Ansible, Docker, Terraform

Database

PostgreSQL, Pinecone, Airtable,
ChromaDB, SQL

CERTIFICATIONS

Deep Learning with Pytorch Specialization

Feb 2026

Coursera & DeepLearning.ai

Reinforcement Learning Specialization

Jan 2026

Coursera & University of Alberta

ROS2 For Beginners and Advanced

Dec 2025

Udemy

Machine Learning Février 2025 Specialization

DeepLearning.ai

Languages

English

French

PROJECTS

Master Thesis on MARL for 3D Urban Mapping

Sept 2025-Present

Developing **Safe Multi-Agent**

Reinforcement Learning policies for
Unmanned Aerial Vehicle **coverage path**

planning in a **Dec-POMDP** framework
using Lagrangian **MAPPO** and **CBF** to
optimize mapping under energy and safety
constraints.

Multimodal AI Assistant for Tourism | Hackaton

May 2024

Designed a vocal multi-language assistant
for tourism using **Hugging Face** and agents
to provide cultural and logistical guidance
in local languages like Fon and Yoruba.

REFERENCES

Prof. Engelbert Mephu Nguifo

Professor, University of Clermont Auvergne
(UCA) LIMOS Laboratory (UMR 6158) Email:-
engelbert.mephu_nguifo@uca.fr

Dr. Viet Hung Nguyen

Associate Professor, University of Clermont
Auvergne (UCA) LIMOS Laboratory (UMR
6158) Email: viet_hung.nguyen@isima.fr

Interests

Sport

Travel